1-Number of Zeros in a Given Array

Started on	Friday, 4 September 2026, 4:28 PM
State	Finished
Completed on	Friday, 4 September 2026, 5:11 PM
Time taken	42 mins 31 secs

Time taken	42 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Open block dr

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int firstzero(int arr[],int low,int high){
    if(low>high)
        return -1;
    int mid=low+(high-low)/2;
    if(arr[mid]==0&&(mid==0||arr[mid-1]==1)){
        return mid;
    }
    if(arr[mid]==1){
        return firstzero(arr,mid+1,high);
    }
    else{
        return firstzero(arr,low,mid-1);
    }
}
```

```c
int main(){
    int m;
    scanf("%d",&m);
    int arr[m];
    for(int i=0;i<m;i++){
        scanf("%d",&arr[i]);
    }
    int index=firstzero(arr,0,m-1);
    if(index==-1)
        printf("0");
    else
        printf("%d",m-index);
    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1	0	0	✔

	1 1 1 1 1			
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1	2	2	✔

	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0			

Passed all tests! ✔

Correct

4-Two Elements sum to x

Started on	Friday, 4 September 2026, 9:59 PM
State	Finished
Completed on	Saturday, 5 September 2026, 10:49 AM
Time taken	12 hours 49 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int findpair(int arr[],int left,int right,int x)
{
    if(left>=right)
        return 0;
    int sum=arr[left]+arr[right];
    if(sum==x){
        printf("%d\n",arr[left]);
        printf("%d\n",arr[right]);
        return 1;
    }
    if(sum<x){
        return findpair(arr,left+1,right,x);
    }
    return findpair(arr,left,right-1,x);

}
int main(){
    int n,x;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    scanf("%d",&x);
    if(!findpair(arr,0,n-1,x)){
        printf("No\n");
    }
    return 0;

}
```

	Input	Expected	Got	
✓	4 2 4 8 10	4 10	4 10	✓

	8 10 14			
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

3-Finding Floor Value

Started on	Friday, 4 September 2026, 9:34 PM
State	Finished
Completed on	Friday, 4 September 2026, 9:58 PM
Time taken	23 mins 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int findFloor(int arr[],int low,int high,int x){
    int floor=-1;
    while(low<=high){
        int mid=low+(high-low)/2;
        if(arr[mid]==x){
            return arr[mid];
        }
        else if(arr[mid]<x){
            floor=arr[mid];
            low=mid+1;
        }
        else{
            high=mid-1;
```

```
        }
    }
    return floor;
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int x;
    scanf("%d",&x);
    int res=findFloor(arr,0,n-1,x);
    printf("%d",res);
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12	2	2	✔

RAJALAKSHMI ENGINEERING COLLEGE

Malavika M 250701409-2025-CSE

M2

Dashboard My courses

CS23331-DAA-2025-CSE / 2-Majority Element

2-Majority Element

Started on	Sunday, 11 October 2026, 7:47 PM
State	Finished
Completed on	Sunday, 11 October 2026, 8:19 PM
Time taken	32 mins 1 sec

Time taken	32 mins 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int countoccurrance(int arr[],int low,int high,int element){
    int count=0;
    for(int i=low;i<=high;i++){
        if(arr[i]==element){
            count++;
        }
    }
    return count;
}
int majorityelement(int arr[],int low,int high){
    if(low==high){
        return arr[low];
    }
    int mid=low+(high-low)/2;
    int leftmajority=majorityelement(arr,low,mid);
    int rightmajority=majorityelement(arr,mid+1,high);
```

```
    if(leftmaj==rightmaj){
        return leftmaj;
    }
    int leftcount=count(arr,low,high,leftmaj);
    int rightcount=count(arr,low,high,rightmaj);
    int n=high-low+1;
    if(leftcount>n/2){
        return leftmaj;
    }
    if(rightcount>n/2){
        return rightmaj;
    }
    return -1;
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int ans=majority(arr,0,n-1);
    printf("%d\n",ans);
    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

5-Implementation of Quick Sort

Started on	Saturday, 5 September 2026, 10:49 AM
State	Finished
Completed on	Saturday, 5 September 2026, 12:05 PM
Time taken	1 hour 16 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include<stdio.h>
void swap(int *a,int *b){
    int temp=*a;
    *a=*b;
    *b=temp;
}
int partion(int arr[],int low,int high){
    int pivot=arr[high];
    int i=low-1;
    for(int j=low;j<high;j++){
        if(arr[j]<=pivot){
            i++;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i+1],&arr[high]);
    return i+1;
}
void quicksort(int arr[],int low,int high){
    if(low<high){
        int pi=partion(arr,low,high);
        quicksort(arr,low,pi-1);
        quicksort(arr,pi+1,high);
    }
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
```

```
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    quicksort(arr,0,n-1);
    for(int i=0;i<n;i++){
        printf("%d",arr[i]);
        if(i<n-1)
            printf(" ");
    }
    return 0;

}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔